

NAVEEN CARLIN A

AI & Full Stack Engineer | Building Scalable, Intelligent, and Impactful Systems

✉ naveencarlin07@gmail.com ☎ +91 7550276890 🔗 [linkedin.com/in/naveen-carlin-a-b66317411](https://www.linkedin.com/in/naveen-carlin-a-b66317411)

<https://carl-portfolio-77777.web.app/> <https://github.com/carl07-gif>

EDUCATION

Bachelor of Engineering (B.E) – Computer Science and Engineering

Nov/2023 - Present

Mount Zion college of Engineering and Technology, Pudukottai | **CGPA:** 8.06/10

PROFESSIONAL EXPERIENCE

- **Research Intern at Alagappa University**, Science Campus, Karaikudi – Department of Management Information Systems. **Gained Practical experience in MIS** operations, enterprise IT systems, and real-time system deployment within an academic infrastructure.
- **Research Intern at National Institute of Technology Tiruchirappalli** – Department of Computer Science and Engineering (CSE). **Worked on AI Based Multi – agent path planning & Enhancing Multiagent Cooperative Path Finding at Signal – Free Intersections through Search Improved Conflict Based Scenario.**
- **V Dart - Full Stack Developer Intern at Tiruchirappalli.** **Developed an** Employee Reward Management System. Using **React Native with AI-driven analytics for employee nomination**, voting, and automated winner selection. **Implemented real-time vote analysis, role-based access control, and performance evaluation** features for awards such as Star Performer and multiple employee recognition categories.

TECHNICAL SKILLS

- Programming Languages: **Java, Python, JavaScript, SQL, PHP**
- Frameworks & Libraries: **JDBC, Flask, React.js, TensorFlow, PyTorch, OpenCV, YOLOv8**
- **Databases:** MySQL, PostgreSQL
- **AI / Computer Vision:** Object Detection (**YOLOv8**), **OCR** (Number Plate Recognition), Image Pre-processing, Real-Time Video Analytics
- **Tools & Platforms:** Git, Linux, REST APIs, Raspberry Pi, OpenCV, SQL Databases
- **Core Expertise:** Full Stack Development, Java Backend Engineering, MVC Architecture, Database-Centric Application Design, Real-Time AI Pipelines, End-to-End System Integration, Secure & Scalable System Development
- **Other Skills:** API Development, Backend Validation & Transaction Handling, System Optimization, Problem Solving, Technical Documentation

KEY PROJECTS

1. Smart Bus Tracking App

- Developed using android studio and integrated Leaflet API to display the bus location on a map. This system improve efficiency and reduces waiting time by providing real time updates to users.

2. AI-Powered Water Level Monitoring System | *NodeMCU ESP8266, FastAPI, MQTT, SQL, OpenAI*

- *API* Developed an IoT-based water monitoring and motor automation system with AI-driven usage prediction.
- Implemented an MQTT–FastAPI backend with SQL-based analytics and fault detection.

3. Hostel Management Application | *Java, SQL, JDBC, MVC*

- Built a modular hostel management system with MVC architecture, supporting room booking, customer records, staff management, and billing.
- Designed optimized SQL schemas for efficient CRUD operations and transaction handling.

4. **AI-Based Vehicle Detection & Number Plate Logging** | *YOLOv8, TensorFlow, OCR, OpenCV, SQL*
- Developed a real-time vehicle detection system using YOLOv8 and OCR, achieving ~95% plate recognition accuracy.
 - Automated IN/OUT logging with timestamps, storing results in an SQL database.
5. **ARGON – Farmer-to-Customer Marketplace** | *HTML, CSS, JavaScript, SQL* Developed a responsive
- web platform enabling farmers to sell directly to customers.
 - Integrated SQL-based sales analytics to identify demand trends and reduce crop wastage.
6. **AI Based Multiagent path planning & Enhancing Multiagent Cooperative Path Finding at Signal – Free Intersections through Search Improved Conflict Based Scenario.**

CERTIFICATES

NPTEL (SWAYAM) Completed Certificates

- ✓ Completed the **NPTEL (ELITE)** certification in **Data Analytics with Python** from **IIT Roorkee** (Jan–Apr 2025, 12-week course), gaining knowledge in Python-based data analysis and analytical problem-solving through graded assignments and a proctored.
- ✓ Completed the **NPTEL certification in Foundation of Cloud IoT Edge ML** from **IIT Kanpur** (Feb–Apr 2024, 8-week course), gaining foundational knowledge in cloud computing, IoT systems, and edge-based machine learning, evaluated through graded assignments and a proctored examination.
- ✓ Completed the **Advanced Google Analytics certification** from **Google Analytics Academy**, gaining practical knowledge in user behavior analysis, conversion tracking, and data-driven performance measurement.
- ✓ Successfully completed “**Python 101 for Data Science**” (PY0101EN), an **IBM-certified course** offered through Cognitive Class, powered by the **IBM Developer Skills Network**, on 26 April 2024.

AWARDS

- Received **Best Paper Award** for the paper titled “AI Based Smart Bus Tracking System” organized by the Department of AI, CSBS, CSE, IT&EEE, K. Ramakrishnan college of Engineering, Tiruchirappalli, at A one Day Inter Collegiate Technical Symposium – SYMPO-2K26, held on 17 March 2026.
- Received **First Prize in “Node Nexus”** at the National Level Technical Symposium – ESPRIE 2K24, organized by the Department of CSE & IT, Sri Bharathi Engineering College for Women, held on 22 October 2024.
- Received a **Second Prize** for Paper Presentation at the National Level Technical Symposium – TECHFEST 2K25, organized by Sudharsan Engineering College, held on 24 April 2025.
- **Participated oral presentation** in the **12th International conference** on Emerging Trends in Engineering and Technology (ICETET'25) held on 4-5 April 2025 at Pandian Saraswathi Yadav Engineering College.
- Recognized as **Top 25 Finalist** across 1.2 Lakh+ submissions in **Naan Mudhalvan – Niral Thiruvizha Hackathon 2025**, organized by the Government of Tamil Nadu for Animal Intrusion Detection.